



Human-Centered
Intelligent Realities

NordiCHI 2026 Workshop

Human-Centered AI Support for Immersive Data Sensemaking

Bringing together HCI, AI, visualization & cognitive science
researchers

HCAI4IDS | October 3, 2026 | Academill, Åbo Akademi University, Vaasa

Contact: organizers.hcai4ids@gmail.com

SESSION 01

Workshop Introduction

09:30-09:45

Organizers



Veronica Sundstedt

Blekinge Institute of
Technology, Sweden



Chaoming Wang

Blekinge Institute of
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Ilir Jusufi

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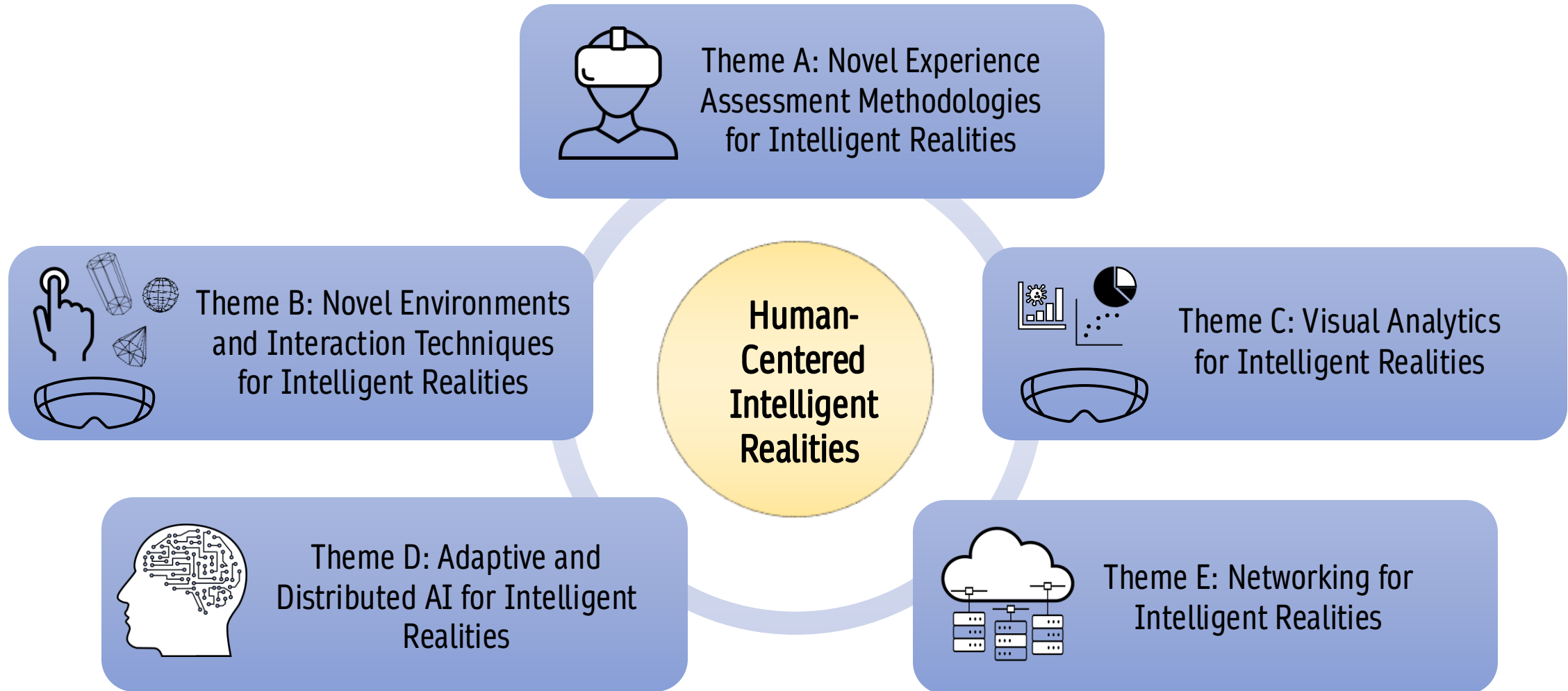
**Christoffer
Johansson**

HealthyGaming / AESF,
Sweden



Ka-i Ip

University of Minnesota –
Twin Cities, USA



HINTS – Theme C

Human-Centered Intelligent Realities



Supporting Data Sensemaking in Immersive Analytics: Exploring AI and Human-Centered Design Perspectives – Chaoming Wang

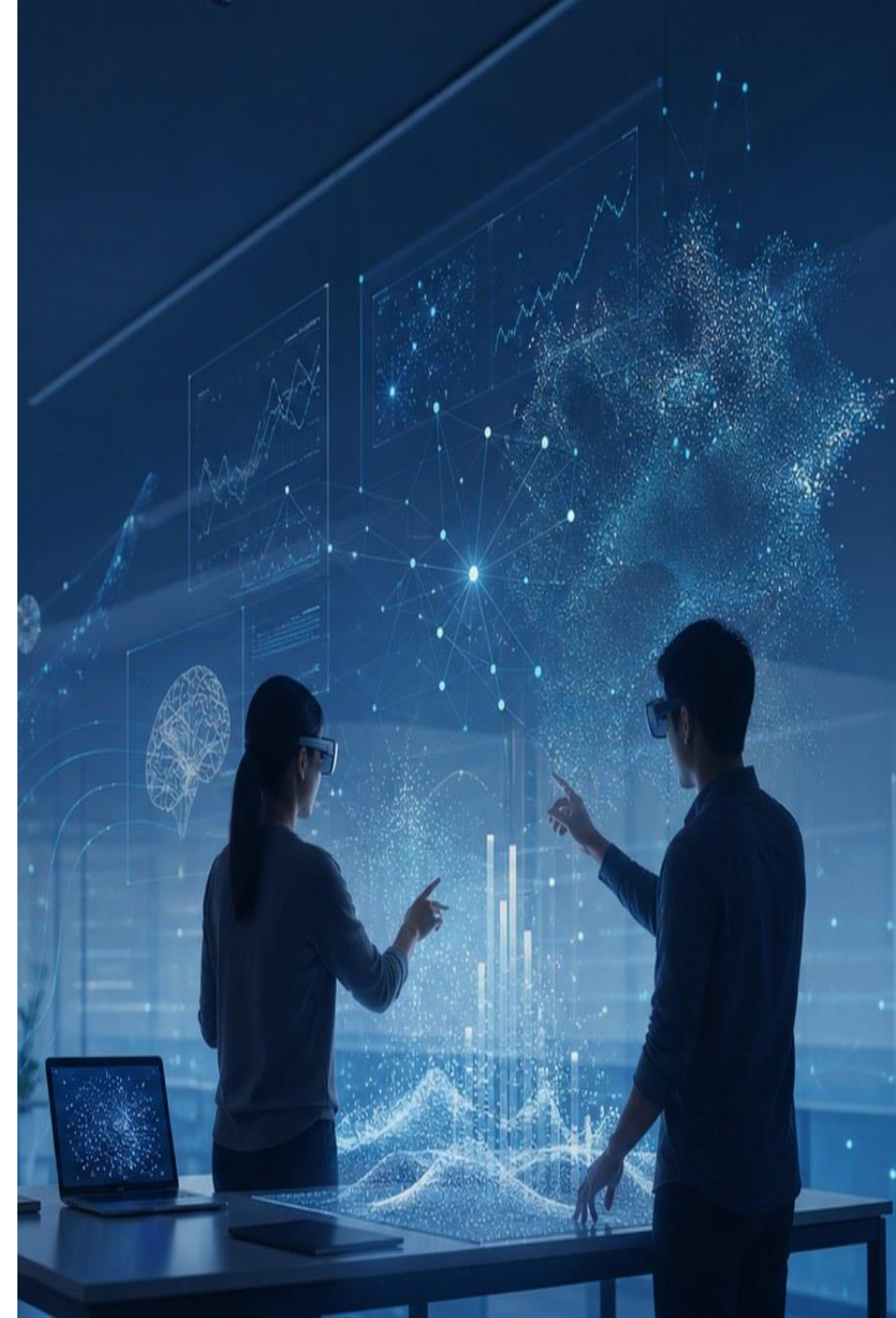
Why this workshop?

Immersive analytics and XR are changing how people explore complex data, reason spatially and collaborate.

Generative AI, adaptive interfaces and multimodal sensing open new ways to summarize, explain, recommend and annotate.

But the integration is fragmented: **how should AI support strengthen human judgment, preserve agency and communicate uncertainty?**

Our stance: AI as a meaningful collaborator in immersive data work — not a replacement for human reasoning.



HCAI4IDS goals

01

Clarify the design space

When should AI guide, explain, adapt, challenge — or step back?

02

Connect communities

HCI, AI, visualization, XR, cognitive science and applied domains

03

Identify human-centered principles

For systems to remain interpretable, accountable, adaptive, and respectful of user agency

04

Discuss evaluation approaches

Trust, collaboration quality, cognitive load, insight generation, user control

05

Develop a shared research agenda

Open challenges, promising methods, future collaboration

Focus areas in immersive data sensemaking



Human-AI collaboration in immersive data sensemaking

Roles, initiative, agency and interaction patterns between users, groups and AI



AI-supported visual analytics in immersive data environments

Explanation, recommendation, summarization, annotation, uncertainty



Multimodal & adaptive interaction to support context-aware AI assistance

Speech, gesture, gaze, physiological signals and interaction traces



Trust, ethics & user control

Transparency, privacy, calibration, accessibility, human oversight



Methods & applications of immersive human-AI sensemaking

Evaluation frameworks, design guidelines, prototypes, domain cases

How today works



01

Present

Short participant presentations and a keynote to seed the discussion



02

Discuss

Theme-based small groups explore challenges and opportunities



03

Synthesize

Consolidate insights into a shared research agenda

EXPECTED OUTCOMES

A shared overview of key challenges and research directions, new collaborations, and a follow-up synthesis article.

Today's agenda

Program

Date: Saturday, October 3, 2026 **Room:** E610 **Time Zone:** GMT+3

Preliminary program: The program is currently being finalized.

09:30 – 09:45	Workshop Introduction Opening remarks, workshop goals, and agenda overview
09:50 – 10:40	Keynote Presentation As we may think next: immersive sense making from the memex to the chatbot Mario Romero Senior Associate Professor Linköping University Hybrid: Room E610 / Join via Zoom
10:50 – 11:05	Paper Presentation 1 10:50–11:05 Supporting the Development of Designers Rather Than Designs: The Role of AI in Design Decisions for Immersive Analytics Presenter: Ingela Rossing
11:05 – 11:25	Coffee Break Coffee break, informal networking, and optional demo showcase
11:25 – 11:40	Paper Presentation 2 11:25–11:40 Where Are We with J.A.R.V.I.S.? An Analysis of Diegetic Human-AI Interaction and Design Implications for Real-World Systems Presenter: Mario Romero
11:40 – 11:50	Break & Setup for Group Work Short transition break and room setup for group work
11:50 – 12:50	Theme-Based Group Work Small-group discussions around workshop themes, opportunities, and challenges
12:50 – 13:00	Future Work and Wrap-Up Workshop synthesis, research agenda, and future collaboration roadmap
13:00 – 14:00	Lunch Networking continues over lunch

SESSION 02

Keynote Presentation

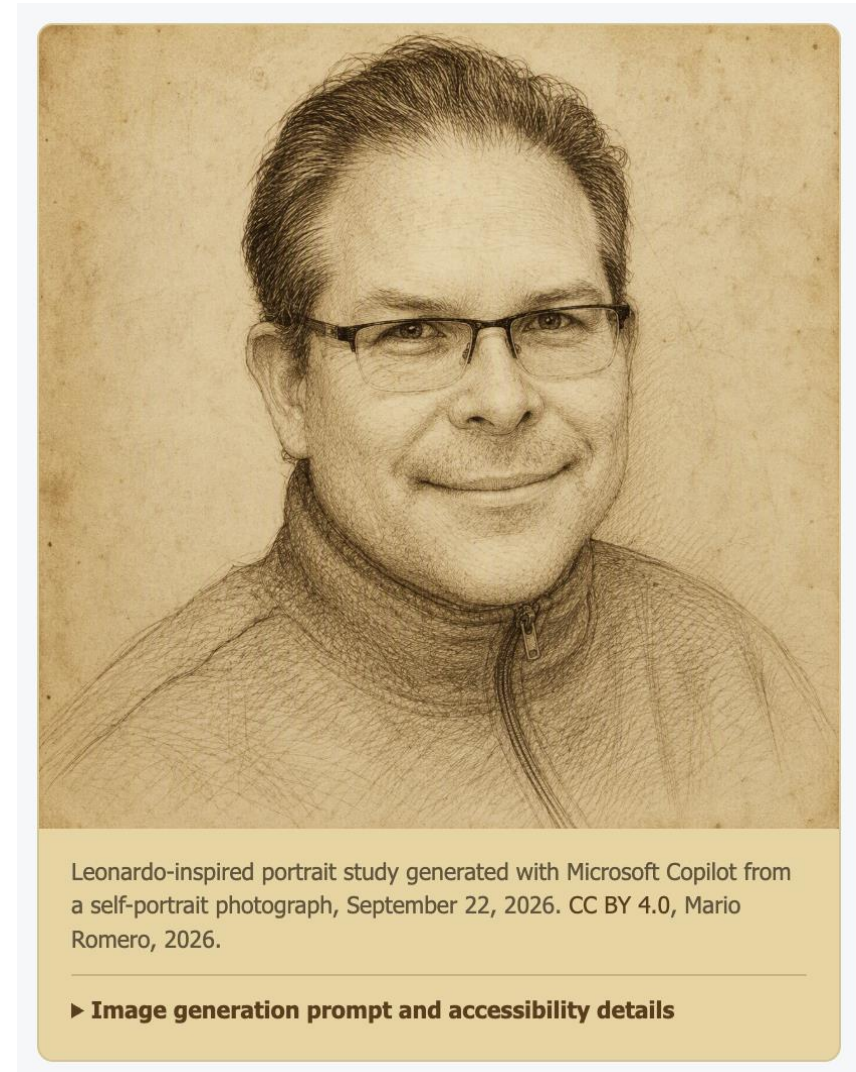
09:50 – 10:40

Invited keynote

Title: As we may think next: immersive sense making from the memex to the chatbot

Mario Romero, Senior Associate Professor,
Linköping University

- National Technical Manager of InfraVis
- His group explores the convergence of AI, immersive visualization, XR, digital twins, and conversational interfaces



SESSION 03

Participant Presentation 1

10:50 – 11:05

Presentation 1

Supporting the Development of Designers Rather Than Designs: The Role of AI in Design Decisions for Immersive Analytics

Ingela L. Rossing and Beat Signer

Web & Information Systems Engineering Lab, Vrije Universiteit Brussel

SESSION 03

Participant Presentation 2

11:25 – 11:40

Presentation 2

Where are we with J.A.R.V.I.S? An analysis of diegetic Human-AI interaction and design implications for real-world systems

Arina Shahbazi, Lonni Besançon, and Mario Romero
Linköping University

SESSION 04

Theme-Based Group Work

11:50 – 12:50

Introduction to the group work

- Four sessions with working cards
- Individual reflections and group work
- Five minute presentation from each group
- Synthesis and sharing of workshop material (no personal information, consent?)

SESSION 05

Future Work and Wrap-Up

12:50 – 13:00

Next steps...

- Synthesis of workshop material as position paper/workshop report (please indicate interest to be involved)
- Research agenda for the next round of HCAI4IDS?
- Other future collaborations?

HCAI4IDS Workshop Feedback

Scan to share your feedback (2–3 minutes)





Thanks !

Contact:

organizers.hcai4ids@gmail.com

LinkedIn Group:

<https://www.linkedin.com/groups/25130030/>

Acknowledgements

Workshop organization is supported by the Knowledge Foundation, Sweden, through the HINTS project (20220068). The invited keynote is supported under contract no. 20210077.